

PRANAY SONI

RESEARCH & DEVELOPMENT LEAD

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PROFESSIONAL PROFILE

Results-oriented seasoned professional with 20 years architecting and deploying production-grade computer / machine vision engineering products for long lasting impact in highly regulated, high-reliability environments (pharma & manufacturing). Adept at taking AI solutions from research to scalable, auditable production deployments, with deep expertise in image processing algorithms, machine learning, generative AI, RAG & knowledge graphs alongside CNNs, ViTs, YOLO, SAM2 and classical CV. Co-founded and scaled a machine vision business making it highly reliable brand, delivering 10+ one of its kind production ready products. Pioneered industry-first auto-teaching AI (80% faster deployment) and anomaly detection. Currently completing Master of Management in AI at Queen's University, combining technical depth with strategic business acumen for enterprise AI initiatives.

CORE COMPETENCIES

Technical: Python • Machine Learning Models (XGBoost, Random Forest, NN) • Generative AI, RAG & GraphRAG systems (LangChain, Neo4j, Ollama) • Deep Learning architectures (CNNs, ViTs, GPT variants, YOLO, SAM2, U-Net) • Zero-shot Anomaly detection, segmentation & object detection pipelines • Model quantization & edge computing • Algorithm design • Sensor fusion • Data preprocessing • Classical & deep learning computer vision • Quality-oriented development • Performance optimization & evaluation. • Data Preprocessing & Model Evaluation, Applied PSI and FSI matrices to assess data quality, distribution shifts, and model deployment readiness.

Leadership & Soft Skills: Cross-functional team leadership & mentoring • Strategic planning • Motivational Leader • Academic & industry collaborations • Stakeholder communication • Customer-centric product development • Research partnership building

AI Governance & Ethics: Developing ethical frameworks for AI-enabled products and services

WORK EXPERIENCE

SPAN INSPECTION SYSTEMS PVT LTD, GANDHINAGAR, INDIA, SEPTEMBER 2015 TO PRESENT (Phased Transition | Fractional / Advisory)

R&D LEAD & CO-FOUNDER

Co-founded and helped scale the company across a decade; now stepped back from day-to-day operations while retaining a minority stake during a phased exit.

- Delivered around ten production vision systems from concept to deployment for Fortune 500 clients, grounding each drive market adoption and address real pain areas. Installed more than 2000 installation across world.
- Co-built the business to over \$6M in annual revenue through strategic technical architecture, product innovation, and customer-centric development.
- Architected and implemented over one million lines of production-quality C++ and Python algorithmic code powering reliable machine vision systems deployed globally.
- Designed and deployed advanced computer vision libraries with object-oriented approach for segmentation, object localization, and anomaly detection, optimized for multi-core CPU and GPU edge devices to achieve sub-100ms processing latencies.
- Led and mentored a cross-functional team of researchers and engineers, consistently meeting project deadlines while raising the technical standard and fostering a collaborative, high-performance culture.
- Built research partnerships with academic institutions that strengthened the R&D pipeline, directly enabling three filed patents and a peer-reviewed publication in an Elsevier journal.
- Developed breakthrough mechanical and opto-mechanical design innovations that enabled machines to outperform competitors by 30% on key metrics, applying detailed observational analysis and nature-inspired principles.

JEKSON VISION, AHMEDABAD, INDIA, APRIL 2006 TO SEPTEMBER 2015

R&D LEAD

- Led development of proprietary image processing libraries tailored for resource-limited environments, eliminating third-party dependencies and optimizing performance on custom hardware platforms.
- Architected complete image processing pipelines spanning pre-processing (noise filtering, histogram equalization) to high-level semantic segmentation and pattern recognition, enhancing reliability and throughput of industrial inspection systems.
- Led an R&D team to deliver multiple market-leading innovations that generated \$4M in new revenue by establishing clear research priorities and fostering technical excellence.
- Developed first-of-its-kind tablet-inspection system on bulk lines through three years of rigorous R&D in software and opto-mechanical systems, driving a 50% boost in company turnover.
- Oversaw the full product lifecycle for multiple systems — from requirements collection through development, deployment, and customer support — ensuring end-to-end quality and client satisfaction.

EDUCATION

Master of Management in AI | Queen's University Smith School of Business, Toronto, ON | 2026 – 2027

Bachelor of Engineering (IT) | Gujarat University, India | 2006

Certification: Advanced Certification Program in Generative AI and Prompt Engineering, IISc Bangalore, India (2024)

PATENTS/PUBLICATIONS

Patents (Approved by Intellectual Property of India): Method and system of automatic inspection of blister pack (595397)

Patents (Provisional): Track & Trace machine with improved changeover (202621010729) • Print & Check machine with improved changeover (202621007538)

Publications: “A novel deep learning pipeline for coronary artery analysis in X-ray angiography,” Engineering Applications of Artificial Intelligence (*Elsevier*, 2025). • “Transforming dataset creation: leveraging generative AI for machine vision applications in data-limited domains,” International Journal of Advances in Electronics and Computer Science.

Awards & Features: Vision Systems Design 2017 Innovators Award for automatic teaching in blister inspection systems. Featured in [Vision Systems Design magazine article](#) on vision system for blister pack inspection.

HACKATHONS

Hack the North 2026 First prize Winner in sponsors category from Embedded Systems with AI by QNX

TECHNICAL SKILLS

Programming & Frameworks: C, C++, Python (production-quality, performance-critical); PyTorch, TensorFlow, OpenCV, ONNX Runtime, OpenVINO, TensorRT, LangChain, FastAPI, Optuna, Neo4j, Ollama.

Computer Vision & Imaging: Classical & deep learning CV pipelines; RGB, X-ray, IR, hyperspectral, 3D (stereo, laser triangulation, ToF); real-time edge optimization; multispectral & sensor fusion.

ADDITIONAL INFORMATION

Languages: English (professional), Hindi, Gujarati

Hobbies & Interests: Ultramarathons & Ironman triathlon, trail running, hiking, photography, physics, robotics, cosmology

Research Interests: Vision language models; Architectural innovations to improve representational capacity of vision models; Novel imaging sensors for capturing semantic information; Bio-inspired and holistic visual understanding beyond pixel-level deep learning.